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Brightloop – Product specializations and Acceptance Criteria

Feature 1: User Registration and Login

1. Purpose

The registration and login system allows users to create accounts and securely access the Brightloop subscription management platform.

2. Functional Requirements

FR1: Users must provide their full name, email address, and password during registration.

FR2: Every registered user must have a unique email address.

FR3: Passwords must contain at least 8 characters.

FR4: During registration, users must be provided with links to the Privacy Policy and Terms & Conditions.

FR5: Users must agree to the Terms & Conditions by selecting a mandatory checkbox before completing registration.

FR6: The Privacy Policy must clearly explain what personal information Brightloop collects, why it is collected, how it is used, and how it is protected.

FR7: The system must prevent registration if the user has not agreed to the Terms & Conditions.

FR8: Users must be able to access the Privacy Policy and Terms & Conditions after registration, such as through the website footer or account settings.

FR9: The system must record the user's acceptance of the Terms & Conditions, including the date and the accepted version.

FR10: Users must be able to log in using their registered email address and password.

FR11: Invalid login credentials must display an appropriate error message.

FR12:Users must agree terms and conditions

FR13: Users must be able to log out securely.

FR14: The system must support different user roles, such as Admin, Staff, and Customer, with appropriate access permissions.

3. Acceptance Criteria

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AC1: A user providing valid registration details can successfully create an account.

AC2: Registration fails when an email address is already registered.

AC3: Passwords shorter than 8 characters are rejected.

AC4: The registration page displays accessible links to the Privacy Policy and Terms & Conditions.

AC5: The Terms & Conditions checkbox is unchecked by default.

AC6: Users cannot complete registration without accepting the Terms & Conditions.

AC7: Clicking the Privacy Policy or Terms & Conditions link opens the correct document.

AC8: Users can access both documents after registration.

AC9: The system correctly records the user's acceptance of the Terms & Conditions.

AC10: Users can log in successfully using valid credentials.

AC11: Incorrect credentials do not allow access to the account.

AC12: After logging out, users cannot access protected pages without logging in again.

AC13: Customers cannot access administrative features.

4. Expected Outcome

Users can securely register, log in, and access only the features permitted for their assigned roles.

Feature 2: Product and Subscription Plan Management

1. Purpose

This feature allows administrators to create and manage products with different variants and subscription plans. Customers can select suitable products and subscribe based on their preferred quantity, price, and billing period.

2. Functional Requirements

FR1: Administrators must be able to create products by entering the product name, description, category, and availability status.

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FR2: Each product must support multiple variants, such as different sizes, quantities, or types.

FR3: Administrators must be able to create multiple subscription plans for each product.

FR4: The system must support four billing periods: daily, weekly, monthly, and yearly.

FR5: Each subscription plan must include a plan name, price, billing period, minimum quantity, and availability dates.

FR6: Administrators must be able to set the minimum quantity required to subscribe to a product.

FR7: Administrators must be able to specify the start and end dates during which a subscription plan is available.

FR8: Customers must be able to view available products, variants, subscription prices, and billing periods before subscribing.

FR9: The system must prevent customers from selecting inactive plans or plans outside their availability dates.

FR10: Administrators must be able to edit product information, variants, prices, and subscription plan details.

FR11: The system must support closing, pausing, and renewing subscriptions according to the configured plan options.

FR12: The system must validate product and plan information and reject invalid entries, such as negative prices, missing product names, or quantities below the required minimum.

FR13: Only authorized administrators must be allowed to create, modify, or deactivate products and subscription plans.

3. Acceptance Criteria

AC1: Administrators can successfully create a product with valid information.

AC2: A single product can have multiple variants.

AC3: Administrators can create daily, weekly, monthly, and yearly subscription plans.

AC4: The system correctly displays the selected product variant, subscription price, and billing period.

AC5: The system prevents subscriptions when the selected quantity is below the minimum requirement.

AC6: Customers cannot subscribe to plans outside their specified availability dates.

AC7: Administrators can successfully update product and plan information.

AC8: The system supports closing, pausing, and renewing subscriptions according to the available options.

AC9: The system rejects invalid prices, missing required information, and invalid quantities.

AC10: Customers cannot access administrative product and plan management features.

4. Expected Outcome

Administrators can efficiently manage products, variants, and subscription plans with flexible billing periods and availability settings. Customers can view available products and select subscriptions according to their preferences, while the system ensures accurate validation and access control.

Feature 3: Quotation to Subscription

1. Purpose

This feature allows businesses to create quotations for customers and convert accepted quotations into subscriptions. It ensures that subscription details, pricing, validity periods, and statuses are properly managed.

2. Functional Requirements

FR1: Authorized users must be able to create quotations for customers by selecting products, variants, subscription plans, and quantities.

FR2: Each quotation must include customer details, selected products, quantities, prices, billing periods, and total amount.

FR3: Administrators or authorized staff must be able to create quotation templates for frequently offered subscription packages.

FR4: Each quotation must have a validity period specifying the number of days for which the offer remains valid.

FR5: The system must prevent expired quotations from being confirmed without renewal or an updated quotation.

FR6: Every subscription must have a unique subscription number for identification and tracking.

FR7: A subscription must support multiple line items, allowing different products or variants to be included.

FR8: The system must support the following subscription statuses:

- Draft
- Quotation
- Confirmed
- Active
- Closed

FR9: Authorized users must be able to convert an accepted quotation into a confirmed subscription.

FR10: The system must update subscription statuses according to the subscription lifecycle.

FR11: Customers must be able to view their own quotation and subscription details.

FR12: The system must prevent unauthorized users from modifying or confirming subscriptions.

3. Acceptance Criteria

AC1: Authorized users can successfully create quotations with valid customer and product information.

AC2: The system correctly displays quotation details, including prices, quantities, and billing periods.

AC3: Quotation templates can be created and reused.

AC4: Each quotation has a clearly defined validity period.

AC5: Expired quotations cannot be confirmed without being renewed or updated.

AC6: Every subscription receives a unique subscription number.

AC7: Multiple products or variants can be included in a single subscription.

AC8: The system correctly displays Draft, Quotation, Confirmed, Active, and Closed statuses.

AC9: Accepted quotations can be converted into confirmed subscriptions.

AC10: Customers can view only their own quotations and subscriptions.

AC11: Unauthorized users cannot modify subscription details.

4. Expected Outcome

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The system provides an organized process for creating quotations, managing their validity, and converting accepted quotations into active subscriptions. Each subscription can be tracked using a unique number and clearly defined statuses.

Feature 4: Invoice Generation and Billing Management

1. Purpose

This feature enables businesses to automatically generate and manage invoices for customer subscriptions. It ensures accurate billing, tax calculations, payment tracking, and timely invoice generation according to the selected subscription plan.

2. Functional Requirements

FR1: The system must automatically generate invoices based on active customer subscriptions and their billing periods.

FR2: Each invoice must contain a unique invoice number, customer details, subscription number, product details, quantity, price, invoice date, and due date.

FR3: The system must support daily, weekly, monthly, and yearly billing cycles.

FR4: Applicable taxes must be calculated automatically based on the configured tax rules.

FR5: The system must calculate the total invoice amount, including applicable taxes and discounts.

FR6: The system must support the following invoice statuses:

- Draft
- Confirmed
- Paid

FR7: Authorized users must be able to review and confirm draft invoices before finalization.

FR8: The system must allow authorized users to send invoices to customers.

FR9: Invoices must be available in a printable format.

FR10: The system must prevent duplicate invoice generation for the same subscription and billing period.

FR11: The system must ensure that scheduled billing periods are not skipped.

FR12: Customers must be able to view their own invoices and payment status.

FR13: The system must prevent unauthorized users from modifying invoice information.

FR14: The system must correctly identify paid and outstanding amounts based on recorded payments.

3. Acceptance Criteria

AC1: An invoice is automatically generated when an active subscription reaches its scheduled billing date.

AC2: Each invoice contains the correct customer, product, subscription, and billing information.

AC3: The system generates invoices according to the selected daily, weekly, monthly, or yearly billing cycle.

AC4: Applicable taxes are calculated correctly.

AC5: Discounts and taxes are reflected accurately in the final invoice amount.

AC6: Invoices correctly display Draft, Confirmed, and Paid statuses.

AC7: Authorized users can confirm and send invoices.

AC8: Invoices can be printed with their details clearly displayed.

AC9: The system does not generate duplicate invoices for the same subscription and billing period.

AC10: Scheduled invoices are not skipped, including when a billing process is delayed or restarted.

AC11: Customers can view only their own invoices.

AC12: The system correctly displays paid and outstanding amounts based on recorded payments.

4. Expected Outcome

The system provides accurate and automated subscription billing by generating invoices on schedule, applying relevant taxes and discounts, maintaining invoice statuses, and preventing duplicate or missed billing. Businesses can efficiently manage invoices while customers can access clear and reliable billing information.

Feature 5: Discount Management

1. Purpose

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This feature allows administrators to create and manage discount rules for subscription plans. It ensures that eligible customers receive the correct discounts based on predefined conditions, including discount type, minimum purchase requirements, validity dates, and usage limits.

2. Functional Requirements

FR1: Only administrators must be allowed to create, edit, activate, or deactivate discount rules.

FR2: The system must support two types of discounts:

- Fixed amount discounts
- Percentage-based discounts

FR3: Each discount rule must contain a name, discount type, discount value, validity start date, and expiry date.

FR4: Administrators must be able to specify minimum purchase amounts or quantities required for a discount.

FR5: Administrators must be able to set usage limits for each discount rule.

FR6: The system must automatically apply eligible discounts to the customer's next invoice.

FR7: Expired or inactive discounts must not be applied.

FR8: Discounts that have reached their usage limits must not be applied.

FR9: The system must verify all discount conditions before applying a discount.

FR10: The invoice must clearly display the original amount, discount applied, and final amount payable.

FR11: The system must prevent discounts from reducing the invoice subtotal below zero.

FR12: Only authorized users must be able to modify discount rules.

3. Acceptance Criteria

AC1: Administrators can successfully create fixed amount and percentage-based discount rules.

AC2: The system correctly calculates discounts according to the selected discount type.

AC3: Discounts are applied only when minimum purchase conditions are satisfied.

AC4: Discounts are applied only within their validity dates.

AC5: Discounts cannot be applied after reaching their usage limits.

AC6: Eligible discounts are automatically reflected in the next invoice.

AC7: The invoice correctly displays the original amount, discount amount, and final payable amount.

AC8: Invalid discount values, including negative amounts and percentages above 100%, are rejected.

AC9: Customers and unauthorized staff cannot create or modify discount rules.

AC10: Applying a discount never produces a negative invoice subtotal.

4. Expected Outcome

The system provides a reliable and automated discount management process that allows administrators to configure promotional offers while ensuring accurate calculations, eligibility verification, and proper application to customer invoices.

- Subshare – Sample Data

1. Product Details

The following fictional products are provided for testing product creation, variant selection, and subscription management.

Product ID	product Name	Category	Available variants
P001	Fresh milk	Diary	500 ml,1 litre
P002	Coffee Powder	Beverages	250g,500g
P003	Newspaper	Publications	English,Malayalam
P004	Mealbox	Food Services	Regular,Premium
P005	Cloud storage	Digital Services	50GB,100GB

All five products are initially marked as active for testing purposes.

2. Subscription Plan Details

The following six fictional subscription plans are prepared for testing Brightloop's product and subscription management features.

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Plan ID| Product| Variant| Plan Name| Billing Period| Price (₹)| Minimum Quantity
PL001| Fresh Milk| 500 ml| Daily Milk Plan| Daily| 30| 1
PL002| Fresh Milk| 1 litre| Monthly Milk Plan| Monthly| 1,800| 1
PL003| Coffee Powder| 250 g| Weekly Coffee Plan| Weekly| 250| 1
PL004| Newspaper| English| Monthly News Plan| Monthly| 200| 1
PL005| Meal Box| Regular| Daily Meal Plan| Daily| 120| 2
PL006| Cloud Storage| 100 GB| Annual Cloud Plan| Yearly| 1,999| 1

Plan Availability

- Start Date: 1 November 2026
- End Date: 31 December 2027
- Initial Status: Active
- Pause Option: Enabled
- Renewal Option: Enabled
- Close Option: Enabled

These settings are proposed sample values and must be confirmed with the development team.

Expected System Behaviour

1. Administrators can create and edit these subscription plans.
2. Customers can view active plans and their prices.
3. The system displays the correct billing period.
4. Customers cannot subscribe below the specified minimum quantity.
5. Plans outside their availability dates cannot be selected for new subscriptions.
6. Subscriptions can be paused, renewed, or closed according to the configured options.

3. Tax Rules

Purpose

To define tax rules that allow Brightloop to automatically calculate applicable taxes when generating subscription invoices.

Sample Tax Data

Tax ID| Tax Name| Rate| Status
TAX001| Zero Tax| 0%| Active
TAX002| Sample Tax A| 5%| Active
TAX003| Sample Tax B| 12%| Active
TAX004| Sample Tax C| 18%| Active

These are fictional testing rules and do not represent the actual GST classification of any product.

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Expected System Behaviour

1. Administrators can create and manage tax rules.
2. Each tax rule contains a name, percentage rate, and status.
3. The system applies the tax rule assigned to the product or subscription.
4. Tax is calculated automatically during invoice generation.
5. Invoices clearly display the taxable amount, tax rate, tax amount, and final total.
6. Inactive tax rules cannot be assigned to new subscriptions.
7. The system rejects invalid tax rates.

Acceptance Criteria

AC1: The system correctly calculates tax based on the assigned rate.

AC2: Products assigned a 0% tax rate do not incur additional tax.

AC3: Invoices correctly display the tax amount and final total.

AC4: Negative tax rates and rates above 100% are rejected.

AC5: Only authorized users can modify tax rules.

Expected Outcome

Brightloop calculates taxes consistently and accurately using configured tax rules, reducing manual billing errors.

4. Discount Rules

Purpose

To prepare sample discount rules for testing Brightloop's automatic discount calculations, eligibility conditions, validity periods, and usage limits.

Sample Discount Data

Discount ID| Discount Name| Type| Value| Minimum Purchase
DIS001| Welcome Offer| Percentage| 10%| ₹500
DIS002| Festive Offer| Fixed| ₹150| ₹1,000
DIS003| Premium Saver| Percentage| 15%| ₹2,000
DIS004| Loyalty Reward| Fixed| ₹200| ₹1,500

Validity and Usage Limits

Discount ID| Start Date| Expiry Date| Maximum Uses
DIS001| 01-11-2026| 31-12-2026| 100
DIS002| 01-11-2026| 30-11-2026| 50

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DIS003| 01-11-2026| 31-03-2027| 75
DIS004| 01-11-2026| 31-12-2027| 30

All discounts are initially marked as Active.

Expected System Behaviour

1. The system checks whether the customer meets the minimum purchase requirement.
2. Discounts are applied only within their validity periods.
3. Percentage discounts are calculated using the eligible subscription subtotal.
4. Fixed discounts deduct the specified amount.
5. Discounts cannot be applied after their usage limits are reached.
6. Expired or inactive discounts must be rejected.
7. The invoice displays the original amount, discount, tax, and final payable amount.
8. The system prevents the discounted subtotal from becoming negative.

Acceptance Criteria

AC1: A 10% discount on an eligible ₹1,000 subscription reduces the subtotal to ₹900.

AC2: A fixed ₹150 discount on an eligible ₹1,000 subscription reduces the subtotal to ₹850.

AC3: A discount is rejected if the minimum purchase requirement is not met.

AC4: Expired discounts cannot be applied.

AC5: Discounts cannot exceed their maximum usage limits.

AC6: The system correctly displays the discount amount on the invoice.

Expected Outcome

Brightloop applies eligible discounts accurately while enforcing purchase requirements, validity dates, and usage limits.

5. Sample Customer Data

Purpose

To provide fictional customer records for testing Brightloop's registration, subscription management, billing, and customer access features.

Customer Details

Customer ID| Customer Name| Email Address| Subscription Plan| Status
C001| Ananya Nair| ananya@example.com| PL001| Active

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C002| Rahul Menon| rahul@example.com| PL002| Active
C003| Fathima Ali| fathima@example.com| PL003| Active
C004| Arjun Kumar| arjun@example.com| PL004| Active
C005| Sneha Thomas| sneha@example.com| PL005| Active
C006| Adil Rahman| adil@example.com| PL006| Active
C007| Meera Krishnan| meera@example.com| PL001| Paused
C008| Vishnu Raj| vishnu@example.com| PL002| Active
C009| Aisha Mohammed| aisha@example.com| PL003| Active
C010| Devika Suresh| devika@example.com| PL004| Closed
C011| Nikhil Das| nikhil@example.com| PL005| Active
C012| Hiba Nazar| hiba@example.com| PL006| Active
C013| Amal Joseph| amal@example.com| PL001| Active
C014| Riya Mathew| riya@example.com| PL002| Paused
C015| Farhan Ahmed| farhan@example.com| PL003| Active
C016| Kavya Prasad| kavya@example.com| PL004| Active
C017| Ashwin Nair| ashwin@example.com| PL005| Closed
C018| Sara Khan| sara@example.com| PL006| Active
C019| Gokul Mohan| gokul@example.com| PL001| Active
C020| Diya Varghese| diya@example.com| PL002| Active

Expected System Behaviour

1. Each customer must have a unique customer ID and email address.
2. Customers can be assigned to available subscription plans.
3. Each customer must be able to view only their own subscription and invoice details.
4. Authorized administrators and staff can view customer records according to their assigned permissions.
5. Subscription statuses must be correctly maintained as Active, Paused, or Closed.
6. Closed subscriptions must not generate future recurring invoices.
7. Paused subscriptions must follow the team's agreed billing and renewal rules.

Acceptance Criteria

AC1: All 20 fictional customer records can be created successfully.

AC2: The system rejects duplicate customer email addresses.

AC3: Customers are correctly associated with their assigned subscription plans.

AC4: Subscription status is displayed correctly.

AC5: Customers cannot access another customer's personal or billing information.

AC6: Administrators can view and manage customer records according to their permissions.

Expected Outcome

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Brightloop can manage multiple customers and their subscription details while maintaining accurate records and appropriate access permissions.

6. Edge Cases and Error Handling

1. Purpose

To identify potential errors and unusual situations in the Brightloop Subscription Management System and define how the application should handle them to ensure reliable operation.

2. Registration and Login

EC1: Duplicate Email Registration

Situation: A user attempts to register with an email address that is already registered.

Expected Result: The system rejects the registration and displays an appropriate error message.

EC2: Invalid Password

Situation: A user enters a password that does not meet the required security conditions.

Expected Result: The system rejects the password and displays the password requirements.

EC3: Terms and Conditions Not Accepted

Situation: A user attempts to register without accepting the Terms & Conditions.

Expected Result: Registration is blocked until the required checkbox is selected.

EC4: Unauthorized Access

Situation: A customer attempts to access an administrator-only page.

Expected Result: Access is denied without exposing administrative information.

3. Product and Subscription Plans

EC5: Invalid Subscription Quantity

Situation: A customer selects a quantity below the minimum requirement.

Expected Result: The system prevents subscription creation and displays the minimum quantity.

EC6: Expired Subscription Plan

Situation: A customer attempts to subscribe to a plan after its availability end date.

Expected Result: The system prevents the new subscription.

EC7: Inactive Product

Situation: A customer tries to subscribe to a deactivated product.

Expected Result: The system rejects the request.

4. Quotation and Subscription

EC8: Expired Quotation

Situation: A customer attempts to confirm a quotation after its validity period.

Expected Result: The system requests an updated or renewed quotation.

EC9: Duplicate Subscription Confirmation

Situation: A user attempts to confirm the same quotation twice.

Expected Result: The system prevents duplicate subscription creation.

EC10: Closed Subscription Renewal

Situation: A customer attempts to renew a closed subscription.

Expected Result: The system follows the agreed reactivation policy and does not renew the subscription without authorization.

5. Invoice Generation

EC11: Duplicate Invoice

Situation: The billing process runs twice for the same subscription and billing period.

Expected Result: Only one invoice is created.

EC12: Missed Billing Date

Situation: The automated billing process fails on the scheduled billing date.

Expected Result: The system identifies and processes the missed billing period without creating duplicates.

EC13: Incorrect Tax Configuration

Situation: A subscription has no valid tax rule assigned.

Expected Result: The system flags the configuration issue and prevents an incorrect invoice from being finalized.

EC14: Invoice Already Paid

Situation: A user attempts to mark an already-paid invoice as paid again.

Expected Result: The system prevents duplicate payment recording.

6. Discount Management

EC15: Expired Discount

Situation: A customer attempts to use a discount after its expiry date.

Expected Result: The discount is rejected.

EC16: Minimum Purchase Not Met

Situation: A customer attempts to use a discount without meeting its minimum purchase requirement.

Expected Result: The discount is not applied.

EC17: Discount Usage Limit Reached

Situation: A discount has already reached its maximum number of uses.

Expected Result: The system prevents further use of the discount.

EC18: Discount Exceeds Subscription Amount

Situation: A fixed discount is greater than the eligible subscription subtotal.

Expected Result: The discounted subtotal is limited to zero and cannot become negative.

7. Expected Outcome

Brightloop handles invalid inputs, expired offers, billing errors, unauthorized access, and unusual subscription conditions without creating inconsistent records or incorrect charges.